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标题	Reactivity and Stability of (Hetero)Benzylic Alkenes via the Wittig Olefination Reaction
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摘要（原文节选）

Wittig olefination at hetero-benzylic positions for electron-deficient and electron-rich heterocycles has been studied. The electronic effects of some commonly used protective groups associated with the N-heterocycles were also investigated for alkenes obtained in the context of the widely employed Wittig olefination reaction. It was observed that hetero-benzylic positions of the pyridine, thiophene and furan derivatives were stable after Wittig olefination. Similarly, electron-withdrawing groups (EWGs) attached to ⋯

数据表 · Table 1（4 行，含数值行 4 行）

Preparation of pyrrole derivatives.

Entry	Substrate	Conditions	Product	Yield %
1		CH3NH2, PTSA (few crystals)rt to 125 ° C, 20 h		25%
2		CH3NH2, PTSA (few crystals)rt to 125 ° C, 20 h		98%
3		5.0 equiv m-chloroanilinexylene, PTSArt to 160 ° C, 24 h		73%
4		1.0 equiv n-BuLi, 1.0 equiv MeITHF, 0 ° C to rt, 1 h		11%

数据表 · Table 2（10 行，含数值行 10 行）

Protection of indole and pyrrole derivatives.

Entry	Substrate	Conditions	Product	%Yield
1		(i) 1.5 equiv NaH, THF, 0 ° C, 30 min(ii) 1.5 equiv AcCl, 0 ° C to rt, 3 h		92
2		(i) 1.5 equiv NaH, THF, 0 ° C, 30 min(ii) 1.5 equiv TsCl, 0 ° C to rt, 2 h		62
3		(i) 1.5 equiv NaH, DMF, 0 ° C, 30 min(ii) 1.5 equiv AcCl, 0 ° C to rt, 3 h		84
4		(i) 1.5 equiv NaH, THF, 0 ° C, 30 min(ii) 1.5 equiv TsCl, 0 ° C to rt, 2 h		72
5		(i) 3.0 equiv NaH, THF, 0 ° C, 30 min(ii) 3.0 equiv Boc2O, 0 ° C to rt, 6 h		67
6		(i) 3.0 equiv NaOH, DCE, 0 ° C, 30 min(ii) 3.0 equiv MsCl, 0 ° C to rt, 24 h		64
7		(i) 3.0 equiv NaOH, DCE, 0 ° C, 30 min(ii) 3.0 equiv MsCl, 0 ° C to rt, 15 h		81
8		(i) 1.5 equiv NaH, THF, 0 ° C, 30 min(ii) 1.5 equiv Boc2O, 0 ° C to rt, 3 h		98
9		(i) 1.5 equiv NaH, THF, 0 ° C, 30 min(ii) 1.5 equiv TBS, 0 ° C to rt, 20 h		75
10		(i) 4.0 equiv KOH, DMSO, 0 ° C, 30 min(ii) 1.2 equiv PhCH2Br, 0 ° C to rt, 12 h		90

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